

Exercises

Problems 1–10 are Core Exercises and contain the essential work for this chapter. Problems 11–20 are Challenge Problems at approximately AMC 8 and MATHCOUNTS level. They are optional for students whose main goal is a solid mathematical foundation, but are strongly recommended for students preparing for competitions. All exercises below have been newly written for this book.

Part A. Core Exercises

1. Solve

$$\begin{aligned}x + y &= 11, \\x - y &= 3.\end{aligned}$$

2. Solve

$$\begin{aligned}x &= 2y + 1, \\3x + y &= 17.\end{aligned}$$

3. Solve

$$\begin{aligned}2x + 3y &= 19, \\2x - y &= 7.\end{aligned}$$

4. Solve

$$\begin{aligned}3x + 4y &= 18, \\5x - 2y &= 4.\end{aligned}$$

5. Suppose

$$4x + 3y = 26, \quad 4x - 3y = 14.$$

Find x without finding y first.

6. Suppose

$$2x + 5y = 31, \quad 7x + 9y = 74.$$

Find $5x + 4y$ without first solving for x and y separately.

7. A theater sells 64 tickets. Student tickets cost \$5 and adult tickets cost \$8. The total revenue is \$398. How many student tickets and adult tickets are sold?
8. A farm has chickens and goats. There are 29 animals and 82 legs. How many chickens and how many goats are there?
9. Decide whether the system has one solution, no solution, or infinitely many solutions:

$$2x + 6y = 10, \quad x + 3y = 5.$$

10. Find the value of k for which the system

$$x + 2y = 7, \quad 3x + 6y = k$$

has infinitely many solutions. For what values of k does it have no solution?

Part B. Challenge Problems

11. Positive numbers
- x
- and
- y
- satisfy

$$x + y = 24, \quad \frac{x}{3} + \frac{y}{4} = 7.$$

Find $x - y$.

12. A collection of 50 coins consists only of quarters and dimes and has a total value of \$8.60. How many quarters are there?

13. Two numbers have sum 41. If twice the larger number is 7 more than three times the smaller number, find the product of the two numbers.

14. Suppose

$$3x + 2y = 17, \quad 5x - 4y = 1.$$

Find $8x - 2y$ without first finding x and y separately.

15. The system

$$(k - 1)x + 2y = 6, \quad 4x + (k + 1)y = 9$$

has exactly one solution for most values of k . Find all values of k for which it does not have exactly one solution.

16. A two-digit number has digit sum 11. Reversing its digits decreases the number by 27. Find the number.

17. At a school event, 120 sandwiches are sold. A regular sandwich costs \$4 and a large sandwich costs \$6. After all sandwiches are sold, the money collected is \$588. Later it is discovered that exactly 12 customers who bought large sandwiches were accidentally charged the regular price. How many large sandwiches were actually sold?

18. Three times one number plus twice another is 50. If 4 is added to the first number and 1 is subtracted from the second, the two new numbers are equal. Find the original two numbers.

19. The equations

$$ax + by = 18, \quad bx + ay = 12$$

have the solution $x = 4$, $y = 2$. Find $a + b$ and $a - b$.

20. A student knows that

$$2x + 3y = 19$$

and wants to determine x and y uniquely by adding one more linear equation. Which of the following would *not* be enough, and why?

- (A) $x - y = 1$, (B) $4x + 6y = 38$,
 (C) $3x + 2y = 16$, (D) $y = 3$.

Part C. Beyond the Chapter

These questions are optional. Their purpose is to practice asking questions, deriving formulas, and summarizing ideas.

21. Starting from

$$ax + by = e, \quad cx + dy = f,$$

derive formulas for x and y by elimination. Explain in words why the quantity $ad - bc$ is important.

22. A store sells two products at prices p and q dollars. A total of N items are sold for R dollars. Derive formulas for the number of each product sold. What restrictions must the data satisfy if the answers represent actual numbers of items?
23. Create two different systems that both have $(x, y) = (3, -2)$ as their unique solution. Then create one system with no solution and one with infinitely many solutions. Explain how you designed each system.
24. Write a six-sentence summary of this chapter. Your summary must explain substitution, elimination, how to choose between them, and the three possible numbers of solutions. End by writing one question about systems of equations that you would like to investigate further.

Answer Key

- | | |
|----------------------------|-------------------------------|
| 1. (7, 4) | 2. (5, 2) |
| 3. (5, 3) | 4. (2, 3) |
| 5. 5 | 6. 43 |
| 7. 38 student, 26 adult | 8. 17 chickens, 12 goats |
| 9. infinitely many | 10. $k = 21$; otherwise none |
| 11. 0 | 12. 24 quarters |
| 13. 390 | 14. 18 |
| 15. $k = -3, 3$ | 16. 74 |
| 17. 66 large | 18. 8, 13 |
| 19. $a + b = 5, a - b = 3$ | 20. (B) |

For Problem 20, equation (B) is just twice the original equation, so it adds no new independent information.

For Problem 21, when $ad - bc \neq 0$,

$$x = \frac{de - bf}{ad - bc}, \quad y = \frac{af - ce}{ad - bc}.$$

Problems 22–24 include open-ended explanation and creation. A strong response should clearly identify independent relationships, justify any restrictions, and explain the reasoning rather than merely state an answer.